

Auteur: Yoren Collier
 Studierichting: Informatica
 Oefeningenreeks 5A nr. 6.7

$$BI \int_{-2}^2 2^{3x}$$

$$\text{Stel: } t = 3x \Rightarrow dt = 3dx \Leftrightarrow dx = \frac{dt}{3}$$

Dan worden de grenzen als volgt: ondergrens: $3 \cdot (-2) = -6$; bovengrens: $3 \cdot 2 = 6$

$$\begin{aligned} \Rightarrow \int_{-6}^6 2^t \frac{dt}{3} &= \left[\frac{2^t}{3 \ln(2)} \right]_{-6}^6 \\ &= \frac{2^6}{3 \ln(2)} - \frac{2^{-6}}{3 \ln(2)} = \frac{64}{3 \ln(2)} - \frac{1}{64 \cdot 3 \ln(2)} = \frac{64^2}{192 \ln(2)} - \frac{1}{192 \ln(2)} = \\ &= \frac{4095}{192 \ln(2)} = \frac{1365}{64 \ln(2)} \end{aligned}$$

References